FORTYGUARD

Heat Intelligence Report

Heatmap Metadata

Location:	33.463836999481686, -112.05783000000001
Date and Time:	2026-11-08 12:00
Temperature:	27.7°C (81.86°F)

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Map



Figure 1: Overview Area



Figure 2: Tile Area

Geographic Analysis

1. General Location

- **City, State, Country:** Phoenix, Arizona, United States
- **Geographic Classification:** Urban core of a major metropolitan area situated in an arid inland basin (the Salt River Valley). Phoenix is the anchor of the Phoenix–Mesa–Chandler metropolitan statistical area, one of the largest and fastest-growing metro areas in the U.S.
- **Köppen Climate Zone:** BWh — Hot Desert Climate. Characterized by extremely hot summers, mild winters, very low annual precipitation (~8 inches), intense solar radiation, and large diurnal temperature swings.
- **Historical Temperature Extremes:** The all-time record high for Phoenix is 122°F (June 26, 1990). The all-time record low is 16°F (January 7, 1913). Phoenix regularly exceeds 110°F in summer and has experienced prolonged multi-week heatwaves, including a record 31 consecutive days at or above 110°F in July 2023.
- **Comparison to Climate Normals:** The average high temperature for Phoenix on November 8 is approximately 79–81°F, with an average daily mean near 66–68°F. The provided reading of 81.86°F at noon is consistent with or very slightly above the expected daily high, suggesting a near-normal to marginally warm early November day. This is within typical variability and does not indicate anomalous heat.

Sources:

- Köppen climate classification system
- NOAA/NWS Phoenix (Sky Harbor) climate normals (1991–2020)
- NWS historical weather records for Phoenix, AZ

2. Terrain & Elevation

- **Elevation:** Approximately 1,085 feet above sea level at the central Phoenix coordinates.
- **Terrain Type:** Phoenix sits in the broad, relatively flat Salt River Valley (Valley of the Sun), a low desert basin surrounded by discontinuous mountain ranges and buttes — including the Phoenix Mountains to the north, South Mountain to the south, Camelback Mountain to the northeast, and the Sierra Estrella to the southwest.
- **Heat Retention & Dissipation:** The basin geography inhibits nighttime radiative cooling. Warm air tends to pool in the valley floor, particularly under calm, clear conditions, slowing overnight temperature decline. During the day, the flat terrain receives intense, largely unobstructed solar radiation. Unlike classic cold-air drainage inversions, Phoenix's basin trapping is primarily relevant for retaining daytime heat and pollutants.
- **Slope Orientation:** The valley floor itself is essentially flat with minimal slope. The surrounding mountains have south- and west-facing slopes that receive maximum solar radiation, contributing to local heating of rock surfaces and re-radiation into the valley. However, at the urban core, slope effects are negligible.

Sources:

- USGS elevation data (National Elevation Dataset)
- Physiographic studies of the Basin and Range Province
- Urban climate studies of the Phoenix metropolitan area

3. Proximity to Water Bodies

- **Nearest Significant Water Bodies:**
  - The Salt River flows through the Phoenix metro area but is largely dry (dammed upstream) through central Phoenix. Tempe Town Lake, an artificial reservoir on the Salt River, lies approximately 7–8 miles to the east-southeast.
  - The Gila River is approximately 20 miles to the south, also often dry or minimal flow.
  - Major reservoirs (Canyon Lake, Saguaro Lake, Apache Lake, Roosevelt Lake) are located 40–70 miles to the northeast in the Tonto National Forest.
  - The nearest ocean (Gulf of California/Sea of Cortez) is approximately 180 miles to the southwest.
- **Thermal Moderating Effect:** Effectively negligible. Phoenix is a profoundly continental/inland arid city. The absence of nearby large water bodies means there is virtually no maritime or lacustrine temperature buffering. Humidity is very low, and evaporative cooling from natural water sources is minimal in the urban core.
- **Seasonal Variation:** During the North American Monsoon (typically July–September), moisture advection from the

Gulf of California and Gulf of Mexico can temporarily increase humidity and cloud cover, providing modest short-term cooling via thunderstorms. By November, this moisture source is inactive, and conditions return to dry and clear.

- **Prevailing Wind Direction:** In November, prevailing winds are generally from the east to southeast during the day. These winds pass over dry desert terrain, offering no moisture-related cooling. Occasional Santa Ana–type offshore flow patterns from the northeast bring warm, dry air.

Sources:

- USGS hydrological data for the Salt and Gila Rivers
- NOAA/NWS Phoenix wind climatology
- North American Monsoon research (Arizona State University)

4. Green Spaces & Vegetation

- **Type and Extent:** Phoenix has significant park and preserve areas, including South Mountain Park/Preserve (~16,000 acres, one of the largest municipal parks in the U.S.), Papago Park (~1,500 acres), and the Phoenix Mountains Preserve. However, these preserves consist primarily of native Sonoran Desert vegetation — not dense forest. Within the urban core, green spaces include irrigated parks (Encanto Park, Steele Indian School Park, Margaret T. Hance Park), golf courses, and residential landscaping.
- **Estimated Canopy Coverage:** Phoenix has an estimated urban tree canopy coverage of approximately 9–10%, one of the lowest among major U.S. cities. The urban core has even less canopy cover due to commercial/industrial land use.
- **Vegetation Type:** Native vegetation is predominantly xerophytic — saguaro, palo verde, creosote, mesquite — which is evergreen or semi-deciduous and provides limited canopy shade. Irrigated ornamental vegetation (non-native species such as eucalyptus, ash, palm trees, citrus, and various shade trees) provides more significant shade but requires substantial water inputs. In November, native deciduous species have reduced foliage, though many xerophytic species retain leaves.
- **Urban Greening Initiatives:** The City of Phoenix has adopted a Tree and Shade Master Plan with a goal of achieving 25% canopy and shade coverage by 2030. This includes tree planting programs focused on shade corridors along streets and at transit stops, particularly in environmental justice communities.
- **Estimated Cooling Contribution:** Irrigated parks in Phoenix have been measured at 3–9°F cooler than surrounding built-up areas in surface temperature, though this "park cool island" effect diminishes rapidly beyond 300–500 feet of the park boundary. The cooling effect is most pronounced in summer; in November, the differential is smaller, estimated at ~2–5°F.

Sources:

- City of Phoenix Tree and Shade Master Plan
- Arizona State University urban ecology research (CAP LTER program)
- American Forests urban canopy assessments

5. Land Cover Type

- **Dominant Surface Materials:** Central Phoenix is dominated by impervious surfaces — asphalt roadways, concrete sidewalks, commercial and industrial roofing (flat built-up roofing, modified bitumen, and some TPO/PVC), and parking lots. Residential areas feature a mix of stucco-clad structures on concrete slabs, decomposed granite (DG) landscaping, and limited turf grass. Bare desert soil (caliche, sandy loam) is present at the urban fringe and in undeveloped parcels.
- **Thermal Properties:**
  - **Asphalt:** Very low albedo (~0.05–0.10), extremely high heat absorption. Surface temperatures on Phoenix asphalt can exceed 160°F in summer. Even in November at noon, asphalt surfaces can reach 110–120°F.
  - **Concrete:** Moderate albedo (~0.20–0.35), high thermal mass, stores and slowly re-radiates heat through evening hours.
  - **Desert soil/DG:** Moderate albedo (~0.20–0.30), lower thermal mass than concrete but retains heat effectively due to dryness (no evaporative cooling).
- **Cool Roof and Pavement Programs:** Phoenix has adopted cool-roof provisions in building codes for commercial structures. The city has also participated in pilot programs for cool pavement coatings (e.g., the CoolSeal pilot program by GuardTop, applied to select streets starting in 2020). Studies showed surface temperature reductions of 10–12°F on treated surfaces. However, adoption remains limited relative to total road and parking surface area.

Sources:

- City of Phoenix Heat Response Plan and cool pavement pilot data
- Arizona State University Global Institute of Sustainability urban heat research
- EPA Urban Heat Island mitigation guidelines

6. Building Density & Height

- **Building Density Characterization:** The coordinates place the analysis near central Phoenix, which features a mixed urban fabric:
  - The downtown core (within ~1 mile) includes mid-rise to high-rise commercial and residential towers.
  - Surrounding areas transition to low-rise commercial strips, light industrial zones, and single-family residential neighborhoods.
- **Typical Building Heights:** Downtown Phoenix high-rises range from 10 to 40 stories (tallest: Chase Tower at 483 feet/40 stories). However, the downtown core is compact; the majority of the Phoenix metro consists of 1- to 3-story structures. Building floor-area-ratio (FAR) in the urban core is moderate by national standards — Phoenix is a sprawling, horizontally-developed city.
- **UHI Contribution from Vertical Massing:** Due to the relatively low-rise, spread-out nature of most of Phoenix, the UHI effect is driven more by vast expanses of impervious surface than by canyon-effect heat trapping from tall buildings. In the small downtown core, multi-story buildings do create localized canyon effects.
- **Common Materials:** Stucco over wood or concrete block is the predominant residential material. Commercial structures use concrete, glass curtain walls, and metal panels. Flat roofs with reflective or dark membranes are standard. Desert-adapted colors (tan, beige) on residential exteriors provide moderate albedo (~0.30–0.40).
- **Anthropogenic Heat Sources:** Extremely significant in Phoenix. Air conditioning is nearly universal and runs for much of the year. The waste heat ejected by HVAC systems (particularly rooftop units) is a major contributor to the urban heat island, estimated to add 1–3°F to ambient nighttime temperatures in the metro area. Vehicle density is high — Phoenix is car-dependent with limited public transit. Industrial heat sources include manufacturing, data centers, and power generation facilities.

Sources:

- City of Phoenix zoning and building data
- Research on Phoenix UHI by Arizona State University (B.L. Turner, M. Grossman-Clarke studies)
- U.S. Department of Energy anthropogenic heat flux studies

7. Shadow Coverage

- **Shadow Coverage Assessment:** At the specific date and time (November 8, noon), the sun is at a moderately high but not extreme elevation. Shadow coverage in central Phoenix at this time is limited — the flat terrain and predominantly low-rise building stock produce short shadows near solar noon. Significant shadow coverage exists only immediately adjacent to mid-rise/high-rise structures downtown and under mature tree canopy.
- **Solar Angle Calculation:**
  - At 33.46°N latitude on November 8, the solar declination is approximately -16.5°.
  - Solar noon elevation angle ≈ 90° – (33.46° + 16.5°) ≈ 40.0°. This is a moderately low sun angle, meaning shadows are approximately 1.2 times the height of the object casting them (tan(40°) ≈ 0.84; shadow length = height / 0.84 ≈ 1.19× height).
  - This relatively low angle produces noticeably longer shadows than in summer (when the solar elevation reaches ~80°+), providing more shade on north-facing walls and north sides of structures.
- **Impact on Temperature:** Shaded surfaces in Phoenix are typically 20–30°F cooler than sun-exposed surfaces in summer. In November, with lower solar intensity, the differential is smaller — estimated at ~10–18°F for surface temperatures, translating to a ~2–5°F ambient air temperature reduction in shaded areas versus fully exposed areas at noon.
- **Permanent vs. Seasonal Shade:** Structural shade (buildings, shade structures, parking canopies) is consistent year-round and is the dominant shade source in central Phoenix. Seasonal shade from deciduous trees (ash, mulberry) is reduced in November as leaf drop begins, though many desert-adapted trees (palo verde, mesquite, ironwood) retain foliage. Palms provide minimal shade due to their narrow canopy profile.

Sources:

- Solar geometry calculations based on NOAA Solar Position Calculator methodology
- Arizona State University shade and thermal comfort research (Ariane Middel et al.)
- EPA guidance on shade and surface temperature relationships

8. Urban Geometry & Street Canyon Effects

- **Street Canyon Characteristics:**
  - **Downtown core:** Estimated H/W ratio of 0.5–1.5 along major streets like Central Avenue and Washington Street. These moderate canyons trap some longwave radiation at night, contributing to elevated nighttime temperatures.
  - **Surrounding residential/commercial areas:** H/W ratios are very low, typically 0.1–0.3, as single-story buildings line wide arterial roads (often 5–7 lanes). These open geometries allow maximum solar exposure during the day but also facilitate faster radiative cooling at night.
- **Wind Corridors:** Phoenix's grid street pattern (aligned approximately N-S and E-W) creates potential wind corridors, but the extremely flat terrain and typically light winds (5–10 mph average) limit the convective cooling benefit. The wide boulevards (e.g., Camelback Road, Indian School Road) are open to wind but also open to full solar exposure, creating a trade-off. Nighttime winds are often calm (2–5 mph), reducing ventilation cooling when it is most needed for UHI mitigation.
- **Street Layout:** Phoenix has a highly regular orthogonal grid oriented to cardinal directions. North-south streets receive direct solar exposure on both sides during morning and afternoon. East-west streets have one side in shade much of the day in November (the north-facing facades), which provides some asymmetric cooling benefit. The grid pattern facilitates uniform solar exposure and does not create significant wind funneling effects.
- **Sky View Factor (SVF):**
  - In the downtown high-rise core, SVF is estimated at 0.4–0.6, which moderately restricts nocturnal longwave radiation escape and contributes to the UHI effect at night.
  - In typical suburban Phoenix, SVF is very high (0.85–0.95), reflecting the open, low-rise character. This high SVF allows efficient radiative cooling at night but also maximizes daytime solar heating of surfaces.
  - The net effect: Phoenix's UHI is driven less by radiative trapping (low SVF) and more by the sheer thermal mass of impervious surfaces, absence of moisture, and anthropogenic heat — making it somewhat distinct from UHI patterns in denser, taller cities.

Sources:

- Urban canyon and sky view factor research applied to Phoenix (Oke, T.R. boundary layer climate principles)
- Arizona State University urban morphology and heat studies
- City of Phoenix street design standards and GIS data

Environmental Factors Analysis

1. Air Quality & Atmospheric Opacity

Typical AQI Range & Classification

- For Phoenix, Arizona in early November, the typical AQI ranges from 30–70, classified as **Good to Moderate** under EPA standards. November marks the transition from monsoon season to the cooler, drier months, generally improving air quality compared to summer ozone peaks.
- Occasional inversions in the Salt River Valley can trap pollutants near the surface, temporarily pushing AQI into the upper Moderate range (51–100).

Primary Pollutants of Concern

- **PM10:** A dominant pollutant year-round due to desert dust, construction activity, and unpaved road surfaces in the greater Maricopa County region. Phoenix frequently records elevated PM10 during windstorm (haboob) events, though these are more common in the summer monsoon season.
- **O_3 (ground-level ozone):** The primary summertime pollutant. By November, reduced solar intensity and shorter days substantially decrease photochemical ozone production; however, residual precursors (NO_x, VOCs) from the 4+ million metro population vehicle fleet remain present.
- **PM2.5:** Moderate concern from vehicle emissions, residential wood burning (which begins to increase in late fall/winter evenings), and occasional wildfire smoke transport from distant fires.
- **NO_2 and CO:** Elevated near major highway corridors (I-10, I-17, Loop 101/202) due to heavy traffic volumes. CO historically was a major wintertime concern in Phoenix but has declined with cleaner vehicle technology.

Influence on Local Temperature

- Aerosol loading from PM10 and PM2.5 produces a modest net warming effect at the surface through absorption of longwave (infrared) radiation emitted by warm urban surfaces, especially during nighttime inversions.
- Daytime particulate haze slightly reduces direct solar transmittance (estimated 2–5% reduction in clear-sky GHI), but in Phoenix's predominantly clear November skies, this effect is minor compared to summer haze episodes.
- The net radiative forcing from urban aerosols in the Phoenix basin is weakly positive (warming), as absorbing dust and soot outweigh the scattering-cooling effect of sulfate aerosols in this arid environment.

Seasonal & Industrial Factors

- November sees declining agricultural burning in surrounding areas and reduced biogenic VOC emissions compared to summer.
- Winter inversion events typically begin in late November through February, trapping pollutants in the shallow boundary layer of the Salt River Valley (elevation ~1,085 feet surrounded by higher terrain).

Heat-Air Quality Feedback Loop

- At 81.86°F, temperatures are warm enough to drive moderate photochemical reactions but well below the critical thresholds (>95°F) that cause rapid ozone exceedances. The heat–ozone feedback is therefore weak on this date.
- Higher temperatures increase evaporative VOC emissions from fueling operations and solvents, which are ozone precursors, but the reduced November daylight hours limit overall ozone accumulation.

Sources:

- EPA Air Quality Index (AQI) standards and breakpoint tables
- Maricopa County Air Quality Department monitoring data norms
- EPA National Emissions Inventory for Maricopa County
- Peer-reviewed literature on desert urban aerosol radiative forcing

2. Climate Classification & Historical Weather

Köppen Climate Classification

- Phoenix, AZ is classified as **BWh — Hot Desert Climate** under the Köppen system. Characterized by extremely hot summers, mild winters, very low annual precipitation (~8 inches/year), and intense solar radiation. Aridity dominates the climate profile year-round.

Historical Climate Normals (November 8)

- Based on 1991–2020 NOAA climate normals for Phoenix Sky Harbor International Airport (the primary climate station):
  - **Average daily high (early November):** ~82–84°F
  - **Average daily low (early November):** ~56–58°F
  - **Average daily mean (early November):** ~69–71°F
- The provided average temperature of **81.86°F** is approximately **11–13°F above the normal daily mean** for this date. This value is closer to the normal daily high than the mean, suggesting either: (a) it represents an afternoon reading rather than a true 24-hour average, (b) an anomalously warm day, or (c) a UHI-influenced measurement. If this is truly the daily mean, it would be significantly above normal — consistent with extreme warm anomaly conditions or a climate-warming signal.
- At solar noon (12:00 hours), actual air temperatures on a normal November 8 in Phoenix would be approximately 78–82°F, ascending toward the daily high around 2:00–3:00 PM. The 81.86°F reading at noon is therefore broadly consistent with a near-normal to slightly warm day when interpreted as an instantaneous noon temperature.

Extreme Temperature Records

- **Record high for November 8 at Phoenix:** Estimated ~96–98°F (extreme records for early November approach the upper 90s).
- **Record low for November 8 at Phoenix:** Estimated ~34–37°F.
- Phoenix holds an all-time record high of 122°F (June 2017) and has seen increasing frequency of record-warm nights.

Long-Term Climate Profile & Urban Heat Behavior

- The hot desert climate produces extreme diurnal heating, high solar loads, and minimal cloud-based radiation shielding. Combined with rapid urbanization, this creates one of the most intense UHI environments in North America.
- Low humidity allows efficient daytime radiative heating but also supports strong nighttime radiative cooling in rural surroundings — the urban fabric disrupts this rural cooling, amplifying the UHI especially at night.
- The valley geography (Phoenix sits in the Salt River Valley surrounded by mountain ranges) can trap warm air and pollutants, contributing to enhanced nocturnal warmth.

Sources:

- NOAA 1991–2020 Climate Normals for Phoenix Sky Harbor International Airport (KPHX)
- Köppen climate classification system (Peel et al., 2007 updated methodology)
- NOAA/NWS Phoenix Weather Forecast Office historical records

3. Climate Trends & Projections

Observed Temperature Trends

- Phoenix has experienced a warming rate of approximately **0.5–0.8°F per decade** over the past 50 years, with nighttime minimum temperatures warming faster than daytime highs — a signature of combined UHI intensification and regional climate change.
- The number of days exceeding 110°F has increased notably since the 1990s. The frequency of "excessive heat warnings" issued by NWS Phoenix has risen substantially over the past two decades.
- Heatwave frequency and duration have increased: the average number of consecutive days above 110°F per year has roughly doubled since the 1970s.
- Humidity trends show a slight increase in dewpoint temperatures during the monsoon season (July–September), but fall/winter months remain consistently arid.

Urban Expansion & Land-Use Change

- The Phoenix metropolitan area has been one of the fastest-growing urban regions in the U.S., with the population of Maricopa County growing from ~2.1 million (1990) to ~4.5 million (2024). This growth has converted vast expanses of desert and agricultural land to impervious urban surfaces.
- Estimated: Urban land area in the Phoenix metro has expanded by 40–50% since 1990, replacing native desert (which has moderate albedo from sandy soils) with lower-albedo asphalt and dark roofing — directly intensifying the UHI.
- Loss of irrigated agricultural land on the urban fringe reduces localized evapotranspiration cooling that historically tempered suburban temperatures.

Climate Projections

- **Near-term (2030s):** Under RCP4.5/SSP2-4.5 scenarios, Phoenix is projected to warm an additional 1.5–2.5°F in mean annual temperature relative to the 2000–2020 baseline. Summer extreme heat days (>110°F) projected to increase by 15–25 days per year.
- **Mid-century (2050s):** Projections indicate 3–5°F of additional warming. The number of days exceeding 100°F could reach 150+ per year (currently ~110). Nighttime temperatures may remain above 90°F for extended periods during summer.
- Precipitation projections suggest drying in winter/spring with uncertain changes in monsoon intensity — net effect likely reduced soil moisture availability and increased drought stress.

Cooling Degree Day (CDD) Trends

- Phoenix already has among the highest CDD totals in the U.S. (~4,000–4,500 CDD annually, base 65°F). This has been increasing at an estimated 50–100 CDD per decade, reflecting both warming temperatures and extended warm seasons.
- Rising CDD directly increases residential and commercial cooling energy demand, generating additional anthropogenic waste heat from HVAC systems, creating a positive feedback loop.

Sources:

- NOAA Climate at a Glance — regional time series for Arizona
- NASA Earth Observatory urban expansion analyses
- Fourth and Fifth U.S. National Climate Assessments (NCA4, NCA5) — Southwest regional chapters
- CMIP6 climate model projections (SSP2-4.5 and SSP5-8.5 scenarios)
- NOAA/NWS cooling degree day data archives

4. Soil Moisture & Permeability

Typical Soil Types

- Soils in the Phoenix metropolitan area are predominantly **aridisols** — characteristic of desert environments. Common types include:
  - **Sandy loam and loamy sand:** Dominant in alluvial fan areas and the Salt River floodplain. High permeability, rapid drainage, low water-holding capacity.
  - **Caliche (calcium carbonate hardpan):** A dense, cemented layer common at shallow depths (often 1–3 feet below surface) across much of the Valley. This layer is nearly impermeable and restricts both infiltration and root penetration.
  - **Clay-rich soils:** Present in some lowland and former agricultural areas, with lower permeability and higher heat retention.

Seasonal Soil Moisture Conditions

- By early November, the summer monsoon season (typically July–September) has ended, and fall precipitation is minimal. Soil moisture levels are **very low**, typically at or near wilting point in undeveloped areas.
- Estimated: Volumetric soil moisture in exposed desert soils around Phoenix in early November is typically 3–8%, well below the level needed for meaningful evaporative cooling.
- Irrigated landscapes (residential lawns, parks, golf courses) maintain artificially elevated soil moisture, creating localized oases of evapotranspiration cooling that can be 5–15°F cooler at the surface than surrounding dry soil or pavement.

Heat Dynamics of Soil

- Dry, sandy desert soils have low thermal conductivity and low thermal mass — they heat rapidly under solar load but also cool relatively quickly after sunset compared to urban materials.
- Caliche layers, being dense and mineral-rich, store more heat than surface sands and can retard subsurface cooling.
- The extreme aridity means **virtually no latent heat flux** from unirrigated soils — nearly all absorbed solar radiation converts to sensible heat, heating the air directly. This is a key driver of the extreme afternoon temperatures characteristic of Phoenix.

Urbanization Effects on Soil

- Estimated: 55–70% of the core Phoenix metro surface is sealed (impervious) by buildings, roads, parking lots, and other structures. This eliminates natural infiltration, destroys evaporative capacity, and replaces low-thermal-mass desert soil with high-thermal-mass artificial surfaces.
- Soil sealing converts the urban surface energy balance strongly toward sensible heat storage and release, the primary mechanism driving the Phoenix UHI.
- Stormwater infrastructure diverts the rare precipitation events away from infiltration, further reducing any soil moisture recharge opportunity.

Groundwater Influence

- The water table in central Phoenix is generally deep (50–150+ feet below ground surface in most areas), with no meaningful influence on surface soil moisture. Groundwater has been heavily pumped for agriculture and municipal use, leading to significant drawdown over the past century.

Sources:

- USDA Natural Resources Conservation Service (NRCS) — Soil Survey of Maricopa County, Arizona
- USGS groundwater monitoring data for the Salt River Valley

- Arizona Department of Water Resources groundwater basin assessments
- Peer-reviewed research on desert urban soil-atmosphere energy exchange

5. Heat Retention Capacity of Surfaces

Dominant Surface Materials

- **Asphalt roadways and parking lots:** Comprise a substantial fraction (~25–35%) of the urban surface in central Phoenix. Dark asphalt has albedo of 0.05–0.15 and very high thermal admittance.
- **Concrete (sidewalks, commercial areas):** Albedo 0.20–0.35, moderate to high thermal mass. Lighter color than asphalt but still a significant heat reservoir.
- **Building rooftops:** A mix of dark composition shingles (albedo ~0.10–0.20), flat commercial roofs (with increasing adoption of cool roof coatings, albedo ~0.60–0.80), and tile roofs (~0.25–0.40).
- **Desert landscaping (xeriscaping):** Decomposed granite, rock mulch, and sparse native vegetation — albedo ~0.20–0.30. Lower thermal mass than pavement but limited evaporative cooling.
- **Irrigated turf and vegetation:** Declining in proportion due to water conservation policies, but still present in older neighborhoods, parks, and golf courses. Very effective at cooling through evapotranspiration.

Surface Temperature Differentials

- Under peak solar load on a day near 82°F ambient:
  - **Asphalt surface:** Estimated 130–145°F
  - **Concrete surface:** Estimated 110–125°F
  - **Bare desert soil:** Estimated 105–120°F
  - **Irrigated grass:** Estimated 80–90°F
  - **Cool roof coating:** Estimated 100–115°F
- The differential between asphalt and irrigated grass can exceed **50–60°F** at peak solar exposure — a dramatic illustration of how surface material dictates microclimate.

Thermal Properties

- **Asphalt:** Thermal admittance ~1,200–1,500 J/(m²·K·s^0.5); stores heat efficiently and releases slowly.
- **Concrete:** Thermal admittance ~1,500–2,000 J/(m²·K·s^0.5); higher thermal mass than asphalt, slower to heat and cool.
- **Desert soil (dry):** Thermal admittance ~600–900 J/(m²·K·s^0.5); relatively fast to heat and cool.
- These properties explain why urban areas retain heat hours after sunset while surrounding desert cools rapidly.

Contribution to UHI Intensity

- The dominance of low-albedo, high-thermal-mass surfaces in Phoenix is the primary driver of its strong UHI, which averages 5–10°F and can exceed 12°F during summer nights.
- In November, with lower solar angles and reduced solar energy input, UHI intensity is reduced but still significant (estimated 3–7°F nighttime differential).

Cooling Infrastructure

- The City of Phoenix has implemented a **Cool Roofs initiative** requiring reflective roofing on new commercial buildings.
- Pilot programs for cool pavements (reflective sealants on asphalt) have been tested in several Phoenix neighborhoods, with measured surface temperature reductions of 10–12°F.
- Increased tree canopy goals: Phoenix has a relatively low tree canopy cover (~9–10%) and has set targets to increase urban shade, particularly along pedestrian corridors.

Sources:

- EPA "Reducing Urban Heat Islands" Compendium
- City of Phoenix Heat Response and Cool Surface Programs
- Lawrence Berkeley National Laboratory — Heat Island Group surface property databases

- Peer-reviewed studies on Phoenix surface energy balance (e.g., Chow et al., Brazel et al., Arizona State University urban climate research)

6. Humidity Levels

Typical Relative Humidity

- November in Phoenix is part of the dry season following the monsoon. At midday (12:00 hours), typical relative humidity is **15–25%** — extremely dry by most standards.
- Morning relative humidity may reach 30–45%, dropping rapidly as temperatures rise. This diurnal swing is characteristic of arid climates with limited moisture sources.

Dew Point Temperature

- Typical range for early November: **30–42°F dew point**. This reflects very dry air with minimal atmospheric moisture.
- At a dew point of ~35°F, the air feels comfortable and crisp. There is no moisture-driven thermal discomfort at these levels.
- Dew points are far below the dangerous heat stress threshold of 75°F, making humidity a non-factor in thermal stress on this date.

Humidity and Perceived Temperature

- With 81.86°F and ~20% relative humidity, the **heat index is essentially equal to the actual temperature** or slightly below it (~80–82°F). In arid conditions, evaporative cooling from skin is highly efficient, and perceived temperature may actually be lower than the thermometer reading.
- This is the primary reason Phoenix at 82°F in November feels significantly more comfortable than 82°F in a humid climate like Houston or Miami.

Seasonal Humidity Patterns

- Peak moisture arrives during the North American Monsoon (July–September), when dewpoints can reach 60–70°F — dramatically increasing heat stress.
- October through June is predominantly very dry, with dewpoints often below 40°F. November is one of the driest months.
- Winter precipitation events associated with Pacific storms (November–March) can temporarily raise humidity, but these are infrequent and short-lived.

Nocturnal Cooling Implications

- Very low humidity is favorable for nocturnal radiative cooling — dry air is more transparent to outgoing longwave radiation, allowing surfaces to cool more effectively.
- In November, this means rural and suburban areas with clear skies can cool 25–35°F overnight — excellent for thermal recovery. Urban areas, however, fail to benefit fully due to thermal mass and sky-view factor limitations from buildings.

Sources:

- NOAA/NWS Phoenix Climate Data — monthly humidity normals
- NWS Heat Index calculation methodology
- Arizona State Climate Office — seasonal moisture climatology

7. Solar Radiation Intensity

Global Horizontal Irradiance (GHI)

- Phoenix receives among the highest solar irradiance in the U.S. On November 8, with clear skies, the **peak GHI at solar noon is estimated at 650–750 W/m²**.
- Daily total GHI for a clear November day in Phoenix: Estimated ~4.5–5.5 kWh/m²/day (substantially lower than the summer peak of ~7.5–8.5 kWh/m²/day but still high by national standards).

Solar Elevation Angle

- At latitude 33.46°N on November 8, the solar declination is approximately -16.5°. The **solar elevation at solar noon is approximately 40–41°**.
- This moderately low angle means:
  - Solar radiation passes through ~1.5 air masses (compared to ~1.0 at zenith), reducing intensity by ~10–15% compared to summer noon.
  - Building shadows are longer, providing more incidental shading in urban canyons — potentially reducing mean radiant temperature at street level.
  - South-facing walls receive substantial direct radiation, becoming significant heat absorbers in the built environment.

Peak Sunlight Hours & Thermal Lag

- Sunrise: ~6:20 AM; Sunset: ~5:30 PM (MST) on November 8 — approximately 11 hours of daylight.
- Peak solar radiation occurs between approximately 11:00 AM–1:00 PM. Maximum air temperature typically follows with a 2–3 hour lag, peaking around 2:00–3:00 PM due to the time required for surface absorption and convective transfer to the air.
- At 12:00 hours, solar input is near its daily maximum, and the 81.86°F temperature is still climbing toward the daily high.

Modifying Factors

- November is one of the clearest months in Phoenix, with typical cloud cover of only 20–30%. Clear-sky conditions predominate, maximizing direct beam radiation.
- Urban canyon geometry in downtown Phoenix (building heights of 100–500 feet along Central Avenue) creates complex shadow patterns that reduce street-level solar exposure during lower sun angles.
- Terrain: The surrounding mountain ranges (South Mountain at ~2,690 feet, Camelback Mountain at ~2,704 feet) do not obstruct midday sun at 33.46°N but can cast early morning or late afternoon shadows on adjacent neighborhoods.

Differential Surface Absorption

- At 700 W/m² incoming solar radiation, asphalt (albedo ~0.10) absorbs ~630 W/m², while a cool roof (albedo ~0.70) absorbs only ~210 W/m². This 3:1 ratio in absorbed energy directly translates to the surface temperature differentials described in Section 5.
- The lower November sun angle increases the importance of vertical surface albedo (building walls) relative to summer, when horizontal surfaces dominate absorption.

Sources:

- NREL National Solar Radiation Database (NSRDB) — TMY data for Phoenix
- NOAA Solar Calculator for solar position
- NASA Surface Meteorology and Solar Energy (SSE) dataset

8. Nighttime Cooling Rates

Diurnal Temperature Range (DTR)

- The typical DTR for Phoenix in November is **25–30°F** — one of the largest diurnal ranges in any major U.S. city, a direct consequence of arid climate and clear skies.
- With a normal high of ~83°F and low of ~57°F, the DTR is approximately 26°F. This large range enables significant nighttime thermal recovery for inhabitants and ecosystems.

Surface Thermal Mass & Urban Geometry Effects

- Urban surfaces (asphalt, concrete, masonry) absorb substantial daytime heat and release it slowly via longwave radiation throughout the night. This stored heat flux can exceed 50–100 W/m² in the hours immediately after sunset in densely built areas.
- Urban canyon geometry (buildings lining streets) reduces the **sky view factor (SVF)** — the fraction of sky visible from

the ground. In downtown Phoenix, SVF can drop to 0.3–0.5, trapping longwave radiation between building faces and slowing radiative cooling.
- Low-density suburban development (which dominates greater Phoenix) has higher SVF and better nighttime cooling than the urban core.

Nighttime Cooling Strength

- In November, Phoenix exhibits **moderately strong nighttime cooling** in rural and suburban areas due to low humidity, clear skies, and high outgoing longwave radiation potential.
- Urban core areas experience **suppressed nighttime cooling** — temperatures may remain 6–10°F warmer than surrounding desert at night, reducing the effective DTR to ~18–22°F in central Phoenix.
- In November, this is less of a health concern than in summer, when nighttime temperatures remain above 90°F in the urban core during extreme events — preventing physiological recovery and increasing heat-related mortality.

Factors Suppressing Cooling

- **Thermal mass:** The dominant suppression mechanism in Phoenix. Extensive asphalt and concrete slowly release stored heat throughout the night.
- **Anthropogenic heat:** HVAC systems, vehicles, commercial operations generate waste heat. In November, residential heating begins in early mornings, adding to nocturnal heat in some areas.
- **Cloud cover:** Infrequent in November (~70–80% clear nights), so this is a minor factor.
- **Humidity:** Minimal suppression effect due to very low moisture content in November.
- **Street canyon trapping:** Significant in the urban core, minimal in sprawling suburban development.

Urban vs. Rural Nighttime Differential

- Estimated November nighttime UHI intensity: **5–8°F**, comparing downtown Phoenix to surrounding Sonoran Desert. This is reduced from the summer peak differential of 8–12+°F due to lower overall heat loading.
- Rural desert locations can cool to 45–50°F on calm, clear November nights while downtown Phoenix may only reach 55–62°F.

Sources:

- NOAA climate normals — diurnal temperature range data for Phoenix (KPHX)
- Arizona State University (ASU) urban climate research — UHI monitoring network data
- Oke et al. (2017), "Urban Climates" — principles of urban energy balance and nocturnal cooling

9. Thermal Comfort Index

Applicable Metric

- The **UTCI (Universal Thermal Climate Index)** is the most appropriate metric for this location and season. Justification: UTCI integrates air temperature, humidity, wind speed, and mean radiant temperature (MRT), and it performs well across the full range of climate conditions — including the hot-arid, high-radiation environment of Phoenix where radiant heat load is a dominant comfort factor.
- The standard NWS Heat Index is less useful here because it loses accuracy below ~40% relative humidity and does not account for radiation or wind, which are critical in Phoenix's outdoor environment.
- WBGT would also be appropriate for occupational heat stress assessment for outdoor workers.

Component Analysis

- **Air temperature:** 81.86°F — warm but moderate for Phoenix.
- **Relative humidity:** Estimated ~18–22% — very low; enhances evaporative cooling from skin.
- **Wind speed:** Typical November midday winds in Phoenix are light — estimated 5–10 mph from variable directions. Provides modest convective cooling.
- **Mean radiant temperature (MRT):** This is the critical variable in Phoenix. In direct sun with ~700 W/m² GHI at a 40° solar elevation, MRT can reach 130–150°F — far exceeding air temperature and significantly increasing thermal stress for exposed individuals. In shade, MRT drops to near air temperature.

Estimated Comfort Classification

- **In direct sun:** Estimated UTCI of approximately 86–93°F (30–34°C), corresponding to **"moderate heat stress"** on the UTCI scale. The high MRT compensates for the low humidity and moderate air temperature.
- **In shade:** Estimated UTCI of approximately 73–79°F (23–26°C), corresponding to **"no thermal stress" to "slight heat stress"** — comfortable conditions.
- This stark sun/shade differential underscores the critical importance of shade provision in Phoenix's urban design.

Heat Stress Risks

- At these November conditions, heat stress risk for the general population is **low to moderate**. However:
  - **Outdoor workers** in construction, landscaping, and agriculture still face moderate risk during sustained midday exposure without shade or hydration — especially given Phoenix's deceptive comfort from dry heat masking fluid loss.
  - **Elderly residents and those experiencing homelessness** — Phoenix has a large unsheltered homeless population (~3,000–4,000 individuals in Maricopa County) who face cumulative thermal exposure.
  - **Children:** Playground surface temperatures can reach 130+°F even at 82°F air temperature, posing burn risk.

Local Mitigation Measures

- Phoenix has invested in **shade structures** along major pedestrian corridors (e.g., Roosevelt Row, downtown light rail stops).
- **Cooling centers** operate primarily during summer months (May–October) but may still be available for vulnerable populations.
- **Misting stations** are deployed at transit stops and outdoor event venues during hot months.
- The City of Phoenix **Office of Heat Response and Mitigation** (established 2021) coordinates heat resilience planning year-round, including tree planting, cool surface deployment, and public education.
- Arizona State University's **SHaDE (Shade, Hydration, and Design for Equity)** research informs local shade policy.

Sources:

- UTCI assessment scale (Bröde et al., 2012; Jendritzky et al., 2012)
- NWS Heat Index and WBGT methodologies
- City of Phoenix Office of Heat Response and Mitigation public reports
- ASU urban climate and thermal comfort research programs

10. Seasonal Variability of Urban Heat Islands

UHI Intensity by Season

- **Summer (June–September):** Maximum UHI intensity. Nighttime urban-rural differential reaches **8–12+°F**. Daytime UHI is paradoxically lower (2–5°F) or even negative in some studies because irrigated urban areas may be cooler than surrounding bare desert during peak insolation ("urban cool island" or "oasis effect").
- **Fall (October–November):** UHI intensity moderates to **5–8°F** at night as overall heat loading decreases. The urban thermal mass still retains meaningful excess heat relative to the rapidly cooling desert.
- **Winter (December–February):** UHI persists at **4–7°F** at night. Reduced solar input means less daytime heat storage, but anthropogenic heat from heating systems and holiday commercial activity contributes. Cool, calm nights with clear skies can maximize the UHI differential.
- **Spring (March–May):** UHI intensifies as solar input increases rapidly. By May, nighttime differentials return to **7–10°F** as the urban fabric begins absorbing and storing intense pre-monsoon heat.

Driving Factors by Season

- **Foliage cycles:** Phoenix has limited deciduous vegetation compared to temperate cities, so foliage changes have modest impact. Some palo verde and mesquite trees lose leaves in winter but provide only partial canopy. Irrigated non-native deciduous trees (e.g., ash, mulberry) drop leaves in fall/winter, reducing shading.
- **Snow/ice:** Not applicable. Snow is exceedingly rare in Phoenix proper (last measurable snow: 1990). The albedo-shift effect that moderates UHI in northern cities has no role here.

- **Monsoon effects:** The July–September monsoon temporarily increases humidity and cloud cover, which can slightly suppress the nighttime UHI by reducing the longwave radiation differential between urban and rural surfaces. However, wet urban surfaces that dry more slowly than desert can actually enhance the UHI through latent heat release.
- **Energy use patterns:** Summer cooling demand is extreme (Phoenix peak electrical load is >7,000 MW on the hottest days). HVAC waste heat (estimated to add 1–3°F to urban temperatures during peak summer) is a meaningful UHI contributor. In November, cooling demand drops sharply, and the HVAC contribution to UHI is minimal.

Urban-Rural Differentials Across Seasons

- ASU's sensor network has documented that the **maximum observed nighttime UHI** in Phoenix can reach 15–18°F during calm, clear summer nights — among the highest in any North American city.
- The **minimum UHI** occurs during windy, overcast winter days when mixing eliminates the differential — sometimes approaching 0°F.
- On this November date, under typical conditions (clear, calm), the nighttime UHI is estimated at **5–8°F**.

Seasonal Adaptation Measures

- **Summer:** Extended cooling center hours, water distribution programs, nighttime outdoor work restrictions (adopted by some jurisdictions), heat emergency declarations.
- **Fall/Winter (current):** Tree planting season — optimal time for establishing new canopy. Cool pavement application projects often occur in fall when temperatures permit proper sealant curing.
- **Year-round:** Building code updates requiring cool roofs, increased shade requirements for parking lots, and urban forestry master plans.
- Planned: The Phoenix **Tree and Shade Master Plan** targets increasing canopy coverage to 25% by 2030 (from ~9–10%), though progress has been slow due to water constraints and the challenge of establishing trees in extreme heat.

Greatest Compound Heat Risk

- **Late June through mid-July** poses the greatest compound heat risk, when:
  - Maximum solar input occurs near the summer solstice.
  - Pre-monsoon humidity begins rising (dewpoints climbing from 40s to 60s°F), increasing heat index values.
  - UHI intensity is near its annual peak.
  - Consecutive nights above 90°F prevent physiological recovery.
  - Cooling energy demand strains the electrical grid, risking outages.
  - Vulnerable populations face the compounding effects of extreme daytime heat, limited nocturnal relief, and rising humidity — the deadliest combination for heat mortality, as evidenced by the record 645 heat-associated deaths in Maricopa County in 2023.
- November, by contrast, represents a **low-risk period** — moderate temperatures, excellent nighttime cooling potential, minimal humidity, and manageable UHI intensity provide a thermally comfortable environment for most of the population.

Sources:

- ASU Urban Climate Research Center — Phoenix UHI monitoring and analysis
- Brazel et al. and Grossman-Clarke et al. — peer-reviewed studies on Phoenix seasonal UHI dynamics
- City of Phoenix Tree and Shade Master Plan
- Maricopa County Department of Public Health — heat-associated death surveillance reports
- EPA Heat Island Effect resources

All respondents consented to the use of their data as outlined in our "Privacy Policy" at the point of collection. Privacy Policy